

A Packing of 8 Four-Spaces in $\mathbb{R}^{11}$

Limin Ge*

October 9, 2026

Abstract

We give 8 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.908876. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.863379, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 8 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.908876 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 8$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.863379, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 8/7 = 2.909091$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N8.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
290887603790881312346953577233091626200640124796440747347338642378202345
288642747318436614226347311464586284556581078609918427435902405818232881
5696831923712500206608943582820123565279027017425
```

denominator:

```
999999999998789582169223288726075352128382700106008431374090492297859835
900434774178400545537287095654686484071889144520530030884713344182795571
555844672676946942297771896359161905309393977859
```

B Coordinates

8 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N8.txt`.

```
-303999382764 113118501801 -301103278103 3027632342197666906503 -346937990944 408059878368 -2960381864896737933 -156806071053 91196296370 20196340562
72609170928 136408258723 438333622373 682552773572 -76766506737 -317251139102 -616873947040 -9549315229700395737 -294438511435 113660775385 227810351333
-23331876790 755122114502 167768713395 -5071633799358136069076 -105053921084 81125460666 -4200025839910052447166 314147306458 -68733700367 445860560357
-10971197995 -67283363324 -257325977979 8019618994272034095830 -263630334506 384682330480 -593067494327861925887 295094982085 407021840248 36044763030
-535370985974 112490475391 -49546775662 -2665933171483017844768 -198178166147 51773435933 133224286766-682977457018 -85742650108 7288886957 -167688003105
235228571395 448458285108 -476549065330 35743458005-410018461518 632801848254 556932329291 11898040520-309276743452 407107470044 192285641256 184575727176
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-185042111194 177840160984 222017892005 28276871480912014251889 -85792521815 151143126839 17448137972 143678885846 183920782212 420797087817 21525709143
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373602617647 -136531175747 218165909096 3670606992384225852767 -22268284568 407226807524 13734321459181695154396 -400409454450 -3470708663 -57831931604
572401748451 198230097526 -42090372410 99736937129 151382869039 301270411367 -175096641949 -411951884380449188035 -38577903051 196464235577 135193165545
33652880106 438514230743 173272718458 622241838103 50620089210 -166203065830 184705076500 232030734195-155484165579 -120416600211 130996671775 370270445529
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269905592528 -93339777229 -365958126908 -43804345832464185970 -168620757357 -7523754613 -621482124708
```